

Data Science Coursework 1 Report

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Module: Data Science

Section 1: Dataset, Hypothesis and Preprocessing

This report is based on a dataset that was obtained from Kaggle, which is a well-known open-source platform used by data scientists to access and share datasets. The dataset used is called *Student Performance Factors* and can be accessed at the following link:

<https://www.kaggle.com/datasets/lainguyn123/student-performance-factors>

The dataset contains around 6600 rows and 20 different attributes related to students' academic performance and their environment. The dataset includes both numerical and categorical data. For example, numerical attributes include *Hours_Studied*, *Attendance*, *Sleep_Hours* and *Exam_Score*, while categorical attributes include *Motivation_Level*, *Family_Income*, *School_Type* and *Gender*. This combination of different types of data makes the dataset suitable for analysing patterns and relationships that affect student performance.

The dataset was selected because it contains a wide range of academic, behavioural and environmental factors, making it suitable for analysing student performance from multiple perspectives. Its size and variety of features make it appropriate for applying data science techniques and drawing meaningful insights.

2.1 Raw Dataset

The raw dataset was first loaded and inspected to understand its structure and content before any preprocessing steps were applied.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Hours_Studi	Attendance	Parental_Invo	Access_to_Res	Extracurricular	Sleep_Hc	Previous_Sc	Motivation	Internet_Tutoring	Family_Incon	Teacher_Qual	School_Type	Peer_Influenc	Physical_Ac	Learning_Dis	Parental_Ec	Distance_fro	Gender	Exam_Score		
2	23	84	Low	High	No	7	73	Low	Yes	0	Low	Medium	Public	Positive	3	No	High School	Near	Male	67	
3	19	64	Low	Medium	No	8	59	Low	Yes	2	Medium	Medium	Public	Negative	4	No	College	Moderate	Female	61	
4	24	98	Medium	Medium	Yes	7	91	Medium	Yes	2	Medium	Medium	Public	Neutral	4	No	Postgradua	Near	Male	74	
5	29	89	Low	Medium	Yes	8	98	Medium	Yes	1	Medium	Medium	Public	Negative	4	No	High School	Moderate	Male	71	
6	19	92	Medium	Medium	Yes	6	65	Medium	Yes	3	Medium	High	Public	Neutral	4	No	College	Near	Female	70	
7	19	88	Medium	Medium	Yes	8	89	Medium	Yes	3	Medium	Medium	Public	Positive	3	No	Postgradua	Near	Male	71	
8	29	84	Medium	Low	Yes	7	68	Low	Yes	1	Low	Medium	Private	Neutral	2	No	High School	Moderate	Male	67	
9	25	78	Low	High	Yes	6	50	Medium	Yes	1	High	High	Public	Negative	2	No	High School	Far	Male	66	
10	17	94	Medium	High	No	6	80	High	Yes	0	Medium	Low	Private	Neutral	1	No	College	Near	Male	69	
11	23	98	Medium	Medium	Yes	8	71	Medium	Yes	0	High	High	Public	Positive	5	No	High School	Moderate	Male	72	
12	17	80	Low	High	No	8	88	Medium	No	4	Medium	High	Private	Neutral	4	No	College	Moderate	Male	68	
13	17	97	Medium	High	Yes	6	87	Low	Yes	2	Low	High	Private	Neutral	2	No	High School	Near	Male	71	
14	21	83	Medium	Medium	Yes	8	97	Low	Yes	2	Medium	Medium	Public	Positive	4	No	High School	Near	Male	70	
15	9	82	Medium	Medium	Yes	8	72	Medium	Yes	2	Medium	Medium	Private	Positive	3	No	Postgradua	Near	Male	66	
16	10	78	Medium	High	Yes	8	74	Medium	Yes	1	Low	Medium	Private	Neutral	4	No	Postgradua	Near	Male	65	
17	17	68	Medium	Medium	No	8	70	Medium	Yes	2	Medium	Medium	Private	Positive	4	No	High School	Near	Female	64	
18	14	60	Medium	Low	Yes	10	65	Low	Yes	0	High	Medium	Private	Positive	3	No	College	Near	Male	60	
19	22	70	Low	Medium	Yes	6	82	Medium	Yes	1	Low	High	Public	Neutral	3	No	High School	Near	Female	65	
20	15	80	Medium	Medium	Yes	9	91	Low	Yes	3	Low	Medium	Public	Positive	2	No	College	Moderate	Female	67	
21	12	75	Medium	High	Yes	7	58	Medium	Yes	3	Medium	Medium	Private	Positive	4	No	College	Near	Male	66	
22	29	78	Medium	Medium	No	5	99	High	Yes	0	High	Medium	Public	Negative	1	No	High School	Moderate	Female	69	
23	19	99	Medium	High	No	6	84	Medium	Yes	1	Medium	High	Public	Neutral	3	No	High School	Near	Male	72	
24	20	74	Medium	High	No	8	89	Low	Yes	1	Medium	Medium	Public	Negative	2	No	College	Moderate	Female	66	
25	11	78	High	Medium	Yes	8	100	High	Yes	1	Low	Medium	Public	Neutral	3	No	High School	Moderate	Male	66	
26	17	65	Low	High	Yes	5	75	Medium	Yes	2	Low	Medium	Public	Positive	3	No	High School	Near	Female	63	
27	21	62	High	Low	Yes	6	54	High	Yes	0	High	High	Public	Positive	3	No	Postgradua	Far	Male	64	
28	13	91	Low	Medium	No	6	90	Medium	Yes	1	High	Medium	Public	Negative	2	Yes	High School	Moderate	Female	65	
29	22	83	High	High	Yes	6	94	Medium	Yes	0	High	Medium	Public	Neutral	2	No	College	Moderate	Male	71	
30	16	90	Medium	Medium	No	4	58	Medium	Yes	1	Low	Medium	Public	Neutral	3	No	High School	Moderate	Female	66	
31	18	66	High	High	No	4	51	Low	Yes	2	Low	Medium	Private	Neutral	2	No	College	Near	Male	64	
32	16	83	Low	Medium	No	6	57	Medium	Yes	0	High	Low	Public	Positive	4	Yes	High School	Near	Female	63	
33	31	70	Medium	High	Yes	7	66	Medium	Yes	1	High	Medium	Public	Negative	3	No	College	Near	Male	69	
34	20	69	Medium	Medium	Yes	8	96	Medium	Yes	2	Medium	High	Public	Positive	3	No	High School	Near	Male	67	
35	14	60	High	Medium	No	5	50	Medium	Yes	2	Medium	Medium	Public	Neutral	3	No	College		Female	61	

Figure 1: Raw dataset sample Excel view

The analysis is carried out from the perspective of a data scientist working for an educational institution. The goal is to better understand student behaviour and identify ways to improve academic outcomes. Based on the dataset, the main idea (hypothesis) is that student performance is influenced by multiple factors such as study habits, environment and personal background, and that analysing these factors can help the institution make better decisions.

To support this idea, three specific problems were identified:

1. Problem 1: Identify students who are at risk of failing, allowing early intervention such as tutoring or additional support.
2. Problem 2: Analyse how different factors such as attendance, study hours, sleep and parental involvement affect student performance.
3. Problem 3: Predict whether a student is likely to pass or fail based on the available data.

These problems are directly related to real-world educational challenges and have practical value, as they can help institutions improve student success rates, optimise resource allocation and make more informed academic decisions.

2.2 Data Preprocessing

Before using the dataset, several preprocessing steps were carried out to make sure the data is clean and usable. First, the dataset was inspected to understand its structure, data types and overall quality. During this step, it was identified that some columns contained missing values, specifically *Teacher_Quality*, *Parental_Education_Level* and *Distance_from_Home*, which could potentially impact the reliability of the analysis if not addressed.

Although the dataset was relatively clean overall, preprocessing was still necessary to ensure consistency, remove potential inaccuracies and prepare the data for further analysis.

Duplicate rows were also checked and removed to ensure that the data is consistent and does not contain repeated information.

Another important step was the creation of a new variable called *Pass*. The original dataset contains *Exam_Score* as a numerical value, but for this analysis it was more useful to convert it into a classification problem. This was done by assigning a value of 1 (pass) to students with a score of 50 or above, and 0 (fail) to those below 50. This transformation enables the dataset to be used effectively for classification techniques and simplifies the prediction task.

2.3 Cleaned Dataset

After preprocessing, the dataset was cleaned and prepared for further analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	
1	Hours_Stu	Attendanc	Parental_I	Access_to	Extracurri	Sleep_Ho	Previous_I	Motivatio	Internet_I	Tutoring_I	Family_In	Teacher_C	School_Ty	Peer_Influ	Physical_A	Learning_I	Parental_I	Distance_I	Gender	Exam_Sco	Pass		
2	23	84	Low	High	No		7	73	Low	Yes		0	Low	Medium	Public	Positive	3	No	High Schoi	Near	Male	67	1
3	19	64	Low	Medium	No		8	59	Low	Yes		2	Medium	Medium	Public	Negative	4	No	College	Moderate	Female	61	1
4	24	98	Medium	Medium	Yes		7	91	Medium	Yes		2	Medium	Medium	Public	Neutral	4	No	Postgradu	Near	Male	74	1
5	29	89	Low	Medium	Yes		8	98	Medium	Yes		1	Medium	Medium	Public	Negative	4	No	High Schoi	Moderate	Male	71	1
6	19	92	Medium	Medium	Yes		6	65	Medium	Yes		3	Medium	High	Public	Neutral	4	No	College	Near	Female	70	1
7	19	88	Medium	Medium	Yes		8	89	Medium	Yes		3	Medium	Medium	Public	Positive	3	No	Postgradu	Near	Male	71	1
8	29	84	Medium	Low	Yes		7	68	Low	Yes		1	Low	Medium	Private	Neutral	2	No	High Schoi	Moderate	Male	67	1
9	25	78	Low	High	Yes		6	50	Medium	Yes		1	High	High	Public	Negative	2	No	High Schoi	Far	Male	66	1
10	17	94	Medium	High	No		6	80	High	Yes		0	Medium	Low	Private	Neutral	1	No	College	Near	Male	69	1
11	23	98	Medium	Medium	Yes		8	71	Medium	Yes		0	High	High	Public	Positive	5	No	High Schoi	Moderate	Male	72	1
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14	21	83	Medium	Medium	Yes		8	97	Low	Yes		2	Medium	Medium	Public	Positive	4	No	High Schoi	Near	Male	70	1
15	9	82	Medium	Medium	Yes		8	72	Medium	Yes		2	Medium	Medium	Private	Positive	3	No	Postgradu	Near	Male	66	1
16	10	78	Medium	High	Yes		8	74	Medium	Yes		1	Low	Medium	Private	Neutral	4	No	Postgradu	Near	Male	65	1
17	17	68	Medium	Medium	No		8	70	Medium	Yes		2	Medium	Medium	Private	Positive	4	No	High Schoi	Near	Female	64	1
18	14	60	Medium	Low	Yes		10	65	Low	Yes		0	High	Medium	Private	Positive	3	No	College	Near	Male	60	1
19	22	70	Low	Medium	Yes		6	82	Medium	Yes		1	Low	High	Public	Neutral	3	No	High Schoi	Near	Female	65	1
20	15	80	Medium	Medium	Yes		9	91	Low	Yes		3	Low	Medium	Public	Positive	2	No	College	Moderate	Female	67	1
21	12	75	Medium	High	Yes		7	58	Medium	Yes		3	Medium	Medium	Private	Positive	4	No	College	Near	Male	66	1
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34	20	69	Medium	Medium	Yes		8	96	Medium	Yes		2	Medium	High	Public	Positive	3	No	High Schoi	Near	Male	67	1
35	14	60	High	Medium	No		5	50	Medium	Yes		2	Medium	Medium	Public	Neutral	3	No	College		Female	61	1

Figure 2: Cleaned dataset sample showing Pass variable

Overall, these preprocessing steps ensured that the dataset is structured, consistent and suitable for applying machine learning techniques in later stages.

Section 2: Proposed Technique

For this analysis, a classification technique is required because the goal is to predict whether a student will pass or fail. Based on the preprocessing step, a new variable called *Pass* was created, which converts the numerical exam score into a binary outcome. Because of this, the problem becomes a classification problem rather than a regression one. Among the techniques covered in the module, Naive Bayes was selected as the most suitable method for this task.

Naive Bayes is a supervised learning algorithm that is commonly used for classification problems. It works based on probability and applies Bayes' Theorem to predict the likelihood of a certain outcome. The key idea behind Naive Bayes is that it assumes all features are independent from each other, even though in real life this is not always true. Despite this assumption being quite simple, the method is known to perform well in many real-world situations, especially when dealing with categorical data.

The dataset used in this report contains a mix of numerical and categorical variables, such as *Motivation_Level*, *Family_Income*, *School_Type* and *Parental_Involvement*.

A significant portion of the dataset consists of categorical features, which further supports the choice of Naive Bayes, as the algorithm is particularly effective when working with such data types.

In addition, the model is relatively simple to understand and implement, which makes it a practical choice for this stage of the coursework, where the focus is on selecting and justifying an appropriate method rather than implementing it.

Another reason for choosing Naive Bayes is its efficiency. The algorithm requires relatively low computational power and can handle large datasets without significant performance issues. Since the dataset contains over 6000 records, this makes Naive Bayes a reasonable option.

Although other methods such as K-Nearest Neighbours or Linear Regression could also be considered, they are less suitable for this specific problem. Linear Regression is more appropriate for predicting continuous values, such as exam scores directly, rather than classifying outcomes into categories. K-Nearest Neighbours, while also suitable for classification, can become computationally expensive with larger datasets and is more sensitive to feature scaling. In comparison, Naive Bayes provides a simpler and more efficient solution for this type of problem.

In conclusion, Naive Bayes was chosen because it fits the classification nature of the problem, works well with the type of data available, and provides a simple yet effective approach to predicting whether a student will pass or fail. The actual implementation of this technique will be carried out in the next stage of the coursework.